DRILL 24

NFPA 1001 Level 2 (JPR 5.4.1) / NFPA 1072

VEHICLE EXTRICATION / HAZMAT

VEHICLE BATTERY ISOLATION – 12V SYSTEMS & ELECTRIC VEHICLES (EV)

⏱️ **Time:** 45–60 min | 🚒 **Apparatus:** Rescue / Engine | 🦺 **PPE:** Bunker Gear, Safety Glasses & Work/Extrication Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Battery Isolation Team

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to safely locate and isolate conventional **12-Volt vehicle batteries** to disarm airbag and electrical fire hazards, and recognize modern **Electric Vehicle (EV) and Hybrid** safety protocols: locating emergency cut loops/first responder disconnects, identifying **bright orange high-voltage cabling**, and understanding high-voltage thermal runaway hazards.

📦 TOOLS & APPARATUS SETUP

- Training vehicle or modern vehicle (volunteer or municipal truck).
- Hand tool battery isolation kit: 10mm/1/2" ratchet wrench, wire cutters/cable shears, electrical tape.
- Emergency Response Guide (ERG) or modern vehicle extrication field guide app (e.g., Euro NCAP / NFPA EV field guide).
- Thermal Imaging Camera (TIC).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE HIGH-VOLTAGE CARDINAL RULE: NEVER CUT, PINCH, OR TOUCH BRIGHT ORANGE CABLES!** High-voltage cables in EVs carry 400 to 800+ Volts DC—instantly fatal.

⚠️ **Airbag Capacitor Bleed Time:** Disconnecting the 12V battery does **not** instantly disarm airbags! Capacitors store backup power for **up to 5 to 10 minutes** after battery isolation. Treat all undeployed airbags as loaded weapons.

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone points a cutting tool near orange cables, or if an EV battery pack shows signs of off-gassing, hissing, or thermal runaway.**

PHASE 1: CONVENTIONAL 12-VOLT BATTERY ISOLATION (15 MINUTES) — SLOW DRY RUN

1 Why Isolate 12V First?

- Eliminates electrical ignition sources around leaking gasoline/antifreeze.
- Cuts power to undeployed Supplemental Restraint Systems (SRS airbags, seatbelt pretensioners).
- Shuts down high-voltage contactors in electric vehicles (EVs require 12V to energize high-voltage relays!).

2 The Disconnect Sequence:

- Pop hood or trunk (look for auxiliary battery locations in trunk or under rear seats).
- Identify terminals: **Negative (-)** (usually black or bare chassis ground) and **Positive (+)** (usually red or plastic cover).
- **Rule: ALWAYS DISCONNECT NEGATIVE (-) FIRST!**
- If your wrench touches the metal car chassis while loosening the negative terminal, nothing happens.
- If you touch the positive terminal first and hit the chassis with your metal wrench, you cause a violent high-amp short circuit, sparks, and battery explosion!
- If cutting cables: Make **two cuts** removing a 2-inch (5cm) chunk of the negative cable so the springy wire ends cannot accidentally spring back and re-contact!

DRILL 24: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication / Hazmat • Practical Application

PHASE 2: ELECTRIC VEHICLE (EV) IDENTIFICATION & HIGH-VOLTAGE SYSTEMS (15 MINUTES) — BUDDY COACHING

1. **Identify the EV/Hybrid:**
   - Badges (Hybrid, EV, PHEV, BEV, Lightning, Tesla, Mach-E).
   - Lack of tailpipes / exhaust system.
   - Exterior charging port doors.
   - Large high-voltage battery casing bolted beneath the floor pan.
2. **Visualizing the Orange Cables:**
   - Locate the inverter, electric drive motors, and high-voltage conduit.
   - Reinforce: All 400V–800V cables are coded in **BRIGHT ORANGE**. They are completely off-limits to extrication tools.

PHASE 3: EV FIRST RESPONDER CUT LOOPS & SERVICE PLUGS (20 MINUTES) — TIMED / STRESS

1. **The Emergency Responder Cut Loop:**
   - Most modern EVs feature a low-voltage **First Responder Cut Loop** or emergency disconnect plug (commonly located under the front hood, near the fuse box, or behind the rear trunk trim).
   - Look for the bright colored tag with a scissors icon or pull-loop.
   - Cutting this low-voltage loop immediately opens the internal high-voltage contactors, trapping high voltage securely inside the sealed battery casing.
2. **EV Fire & Submersion Precautions:**
   - If an EV battery is damaged or burning, water requirements are massive (minimum **10,000 to 20,000 Litres** / 3,000–5,000 gallons) to cool the battery cell casing.
   - Use a TIC to monitor battery casing temperature continuously from a safe distance.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Cutting positive battery terminal first:* A metal tool bridging the positive post to the fender creates a blinding arc flash and molten metal spatter. Always isolate the negative ground first!
- ❌ *Cutting into B-pillars without checking for seatbelt inflators:* Modern B-pillars contain pyrotechnic gas cylinders. Cutting through an inflator causes an explosive metal rupture. Peel plastic trim back to inspect before cutting!
- ❌ *Assuming an EV is off because the engine is silent:* An EV can be fully energized and in gear without making any sound. Always chock wheels and remove the key fob 50+ feet away from the vehicle.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Where is the low-voltage cut loop located on our municipal hybrid/EVs?", "Why must we wait 5–10 minutes after disconnecting 12V before cutting near airbags?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hand tools (10mm wrenches, cable cutters) wiped clean and stowed in extrication kit.
- ☐ Training vehicle hoods closed.
- ☐ Vehicle emergency guidebooks returned to command cab.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS